DBMS LAB ASSIGNMENT-2

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QUESTION 1: CREATE EMP1 TABLE

```
CREATE TABLE EMP1 (
    ID INT PRIMARY KEY,
    Name VARCHAR(10),
    Basic DECIMAL(6,2),
    Designation VARCHAR(10),
    Age INT
);

DESC EMP1;
```

OUTPUT:

Field	Type	Null	Key	Default	Extra
ID	int	NO	PRI		
Name	varchar(10)	YES			
Basic	decimal(6,2)	YES			
Designation	varchar(10)	YES			
Age	int	YES			

Result: Structure of EMP1 table is displayed.

QUESTION 2: CHANGE DATA TYPE OF BASIC

```
ALTER TABLE EMP1
MODIFY Basic INT;

DESC EMP1;
```

OUTPUT:

Field	Type	Null	Key	Default	Extra
ID	int	NO	PRI		
Name	varchar(10)	YES			
Basic	int	YES			
Designation	varchar(10)	YES			
Age	int	YES			

Result: Basic column data type is changed from DECIMAL(6,2) to INT.

QUESTION 3: CHANGE NAME SIZE

```
ALTER TABLE EMP1
MODIFY Name VARCHAR(15);

DESC EMP1;
```

OUTPUT:

Field	Type	Null	Key	Default	Extra
ID	int	NO	PRI		
Name	varchar(15)	YES			
Basic	int	YES			
Designation	varchar(10)	YES			
Age	int	YES			

Result: Name column size is changed from VARCHAR(10) to VARCHAR(15).

QUESTION 4: CREATE EMP_trainee TABLE

```
CREATE TABLE EMP_trainee (
    Emp_id INT,
    Name VARCHAR(15),
    Basic INT,
    Designation VARCHAR(10),
    Age INT
);
```

```
DESC EMP_trainee;
```

OUTPUT:

Field	Type	Null	Key	Default	Extra
Emp_id	int	YES			
Name	varchar(15)	YES			
Basic	int	YES			
Designation	varchar(10)	YES			
Age	int	YES			

Result: EMP_trainee table is created with Emp_id in place of ID.

QUESTION 5: INSERT DATA INTO EMP1

```
INSERT INTO EMP1 (ID, Name, Basic, Designation, Age)
VALUES
(1, 'Rohit', 6700, 'Manager', 24),
(2, 'Sunil', 6200, 'Engineer', 27),
(3, 'Payal', 6300, 'Engineer', 25),
(4, 'Kunal', 6700, 'Trainee', 28),
(5, 'Sunita', 6230, 'Trainee', 26),
(6, 'Bimal', 7000, 'Trainee', 25);

SELECT * FROM EMP1;
```

OUTPUT:

ID	Name	Basic	Designation	Age
1	Rohit	6700	Manager	24
2	Sunil	6200	Engineer	27
3	Payal	6300	Engineer	25
4	Kunal	6700	Trainee	28
5	Sunita	6230	Trainee	26
6	Bimal	7000	Trainee	25

Result: 6 rows are inserted into EMP1.

QUESTION 6: INSERT TRAINEE RECORDS

```
INSERT INTO EMP_trainee (Emp_id, Name, Basic, Designation, Age)
SELECT ID, Name, Basic, Designation, Age
FROM EMP1
WHERE Designation = 'Trainee';

SELECT * FROM EMP_trainee;
```

OUTPUT:

Emp_id	Name	Basic	Designation	Age
4	Kunal	6700	Trainee	28
5	Sunita	6230	Trainee	26
6	Bimal	7000	Trainee	25

Result: Only the three records whose Designation is Trainee are copied. Run this INSERT once; if repeating it, clear EMP_trainee first to avoid duplicate rows.

QUESTION 7: ADD SKILLS AND DOJ COLUMNS

```
ALTER TABLE EMP1
ADD Skills VARCHAR(10),
ADD DOJ DATE;

DESC EMP1;

-- Add Skills and DOJ values
UPDATE EMP1 SET Skills = 'Manage',  DOJ = '2020-01-15' WHERE ID = 1;
UPDATE EMP1 SET Skills = 'Develop', DOJ = '2020-02-20' WHERE ID = 2;
UPDATE EMP1 SET Skills = 'Testing', DOJ = '2020-03-10' WHERE ID = 3;
UPDATE EMP1 SET Skills = 'Manage',  DOJ = '2020-04-15' WHERE ID = 4;
UPDATE EMP1 SET Skills = 'Design',  DOJ = '2020-05-20' WHERE ID = 5;
UPDATE EMP1 SET Skills = 'Develop', DOJ = '2020-06-25' WHERE ID = 6;

SELECT * FROM EMP1;
```

OUTPUT:

ID	Name	Basic	Designation	Age	Skills	DOJ
1	Rohit	6700	Manager	24	Manage	2020-01-15
2	Sunil	6200	Engineer	27	Develop	2020-02-20
3	Payal	6300	Engineer	25	Testing	2020-03-10
4	Kunal	6700	Trainee	28	Manage	2020-04-15
5	Sunita	6230	Trainee	26	Design	2020-05-20
6	Bimal	7000	Trainee	25	Develop	2020-06-25

Result: Two columns, Skills and DOJ, are added and values are updated for all six employees.

QUESTION 8: UPDATE MORE THAN ONE ROW IN ONE QUERY

```
-- Increase Basic salary by 500 for all Trainees
SET SQL_SAFE_UPDATES = 0;

UPDATE EMP1
SET Basic = Basic + 500
WHERE Designation = 'Trainee';

-- Turn Safe Update Mode ON again
SET SQL_SAFE_UPDATES = 1;

SELECT * FROM EMP1;
```

OUTPUT:

ID	Name	Basic	Designation	Age	Skills	DOJ
1	Rohit	6700	Manager	24	Manage	2020-01-15
2	Sunil	6200	Engineer	27	Develop	2020-02-20
3	Payal	6300	Engineer	25	Testing	2020-03-10
4	Kunal	7200	Trainee	28	Manage	2020-04-15
5	Sunita	6730	Trainee	26	Design	2020-05-20
6	Bimal	7500	Trainee	25	Develop	2020-06-25

Result: Basic salary is increased by 500 for all employees with Designation = 'Trainee'.

QUESTION 9: RENAME AGE COLUMN

```
ALTER TABLE EMP1
RENAME COLUMN Age TO Age_in_Years;

DESC EMP1;
```

OUTPUT:

Field	Type	Null	Key	Default	Extra
ID	int	NO	PRI		
Name	varchar(15)	YES			
Basic	int	YES			
Designation	varchar(10)	YES			
Age_in_Years	int	YES			
Skills	varchar(10)	YES			
DOJ	date	YES			

Result: Age column is renamed to Age_in_Years.

QUESTION 10: DROP AGE COLUMN FROM EMP_trainee

```
ALTER TABLE EMP_trainee
DROP COLUMN Age;

DESC EMP_trainee;
```

OUTPUT:

Field	Type	Null	Key	Default	Extra
Emp_id	int	YES			
Name	varchar(15)	YES			
Basic	int	YES			
Designation	varchar(10)	YES			

Result: Age column is removed from EMP_trainee.

QUESTION 11: RENAME TABLE EMP1

```
RENAME TABLE EMP1 TO EMP_Mgr_Engr;

SHOW TABLES;
```

OUTPUT:

Tables_in_DBMS_LAB_ASSIGNMENT2
EMP_Mgr_Engr
EMP_trainee

Result: EMP1 is renamed to EMP_Mgr_Engr. Both tables are listed.

QUESTION 12: TRUNCATE EMP_Mgr_Engr TABLE

```
TRUNCATE TABLE EMP_Mgr_Engr;

SELECT * FROM EMP_Mgr_Engr;
```

OUTPUT:

ID	Name	Basic	Designation	Age_in_Years	Skills	DOJ

Result: All records are deleted from EMP_Mgr_Engr. The table structure remains, and the SELECT query returns an empty result set.